1. Вежба
1. Задатак

Никола Шошевић 2514/26

X(mm)	Y(t/ha)
520	3.1
610	3.8
700	[illegible].9
820	5.1
910	5.3
1030	6.0
1120	6.2
1250	5.9
1390	5.1
1510	4.1

$N=10$

$$y(x,w)=\sum_{j=0}^{M} w_j x^j$$

① $$Err(w)=\frac{1}{2N}\sum_{i=1}^{N}(y_i-\overbrace{y(x_i,w)}^{\hat{y}_i})^2$$

$$\hat{y}_i=\sum_{j=0}^{M} w_j x_i^j$$

$$e_i=y_i-\hat{y}_i \Rightarrow e_i=y_i-\sum_{j=0}^{M} w_j x_i^j$$

$e_i=a$ - грешка за појединачно мерење

$$\Downarrow$$

$$\boxed{Err(w)=\frac{1}{2N}\sum_{i=1}^{N}\left(y_i-\sum_{j=0}^{M} w_j x_i^j\right)^2}$$

- Зашто ф-ја одговара критеријуму најмањих квадрата?

- Зато што минимизује суму квадрата разлика између стварних и предвиђених вредности.
Квадрирањем се избегава поништавање позитивних и негативних грешака.

[redacted]

② $$\frac{\partial Err}{\partial w_k}=0 \text{ за } k=0,1,\dots,M$$

$$Err(w)=\frac{1}{2N}\sum_{i=1}^{N}\left(y_i-\sum_{j=0}^{M} w_j x_i^j\right)^2$$

извод по w_k:

$$\frac{\partial}{\partial w_k}(u^2)=2u\frac{\partial u}{\partial w_k}$$

$$\rightarrow \frac{\partial Err}{\partial w_k}=\frac{1}{2N}\sum_{i=1}^{N}2\left(y_i-\sum_{j=0}^{M} w_j x_i^j\right)\frac{\partial}{\partial w_k}\left(y_i-\sum_{j=0}^{M} w_j x_i^j\right)$$

$\rightarrow$

$$\frac{\partial}{\partial w_k}\left(y_i - \sum_{j=0}^{M} w_j x_i^j\right) = -x_i^k$$

$$\frac{\partial Err}{\partial w_k} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \sum_{j=0}^{M} w_j x_i^j\right)\left(-x_i^k\right)$$

$$\Downarrow$$

$$\boxed{\frac{\partial Err}{\partial w_k} = -\frac{1}{N}\sum_{i=1}^{N} x_i^k\left(y_i - \sum_{j=0}^{M} w_j x_i^j\right)}$$

Услов минимума:

$$-\frac{1}{N}\sum_{i=1}^{N} x_i^k\left(y_i - \sum_{j=0}^{M} w_j x_i^j\right) = 0 \quad / \cdot (-N)$$

$$\sum_{i=1}^{N} x_i^k\left(y_i - \sum_{j=0}^{M} w_j x_i^j\right) = 0$$

$$\sum_{i=1}^{N} x_i^k y_i - \sum_{i=1}^{N}\sum_{j=0}^{M} \overset{const.}{w_j}\, x_i^{j+k} = 0$$

$$\sum_{i=1}^{N} x_i^k y_i = \sum_{j=0}^{M} w_j \sum_{i=1}^{N} x_i^{j+k} \Rightarrow \boxed{\sum_{i=1}^{N} x_i^k y_i = \sum_{j=0}^{M} w_j \sum_{i=1}^{N} x_i^{j+k}, \; k = 0, 1, \dots, M}$$

③ $(\phi^T\phi)w = \phi^T y$

$$\sum_{i=1}^{N} x_i^k y_i = \sum_{j=0}^{M} w_j \sum_{i=1}^{N} x_i^{j+k} \quad ; k = 0, 1, \dots, M$$

$$\phi_{ij} = x_i^j$$

$$\phi = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^M \\ 1 & x_2 & x_2^2 & & x_2^M \\ \vdots & & & & \\ 1 & x_N & x_N^2 & & x_N^M \end{bmatrix} ; \quad w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_M \end{bmatrix} ; \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

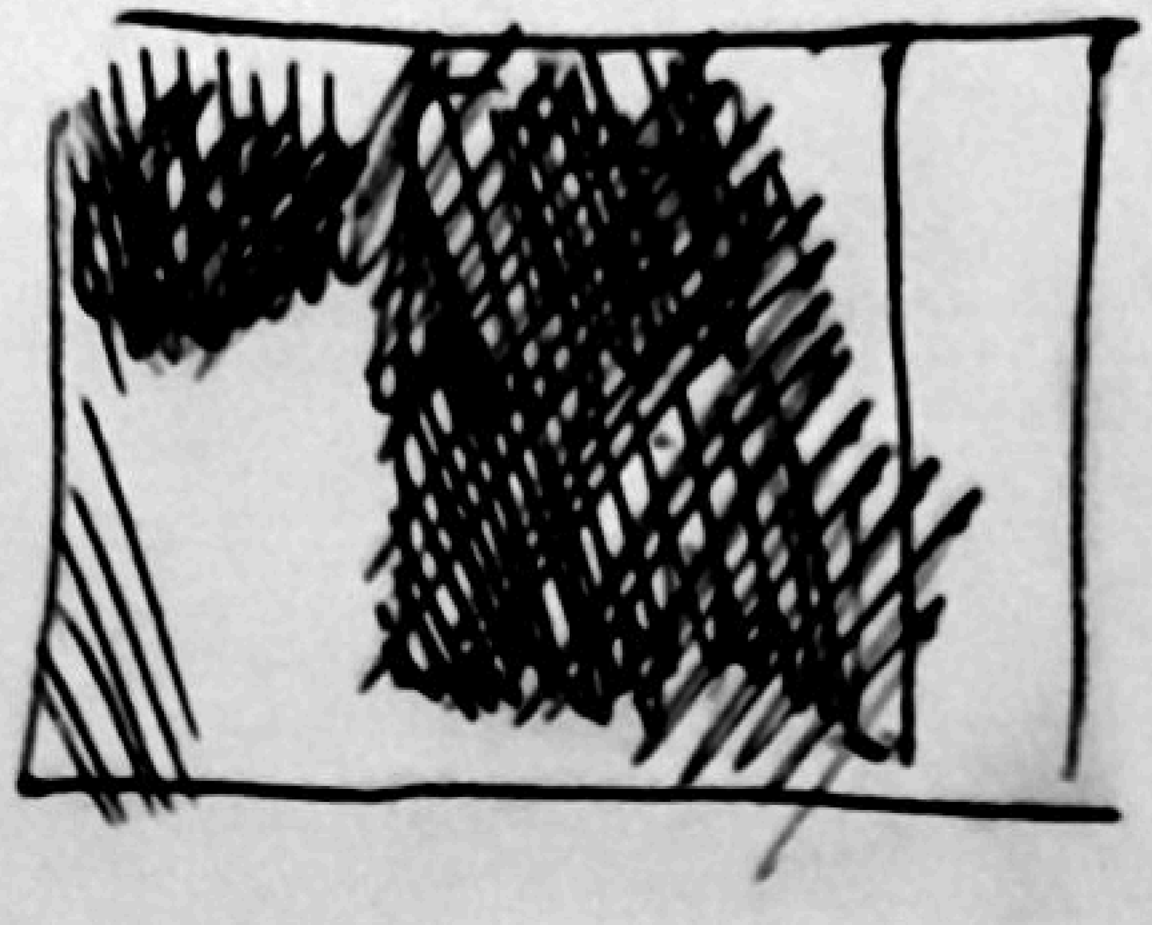
$$\phi^T = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & & x_N \\ x_1^2 & x_2^2 & & x_N^2 \\ \vdots & \vdots & & \\ x_i^M & x_i^M & & x_N^M \end{bmatrix} ; \quad \phi^T\phi = \begin{bmatrix} \Sigma x_i^0 & \Sigma x_i^1 & \cdots & \Sigma x_i^M \\ \Sigma x_i^1 & \Sigma x_i^2 & & \Sigma x_i^{M+1} \\ \vdots & & & \vdots \\ \Sigma x_i^M & \Sigma x_i^{M+1} & & \Sigma x_i^{2M} \end{bmatrix}$$

$$\phi^T y = \begin{bmatrix} \Sigma x_i^0 y_i \\ \Sigma x_i^1 y_i \\ \vdots \\ \Sigma x_i^M y_i \end{bmatrix} ; \quad (\phi^T\phi)w = \phi^T y = \begin{bmatrix} \Sigma x_i^0 & \Sigma x_i^1 & \cdots & \Sigma x_i^M \\ \Sigma x_i^1 & \Sigma x_i^2 & & \Sigma x_i^{M+1} \\ \vdots & & & \vdots \\ \Sigma x_i^M & \Sigma x_i^{M+1} & & \Sigma x_i^{2M} \end{bmatrix} \cdot$$

$$\cdot \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_M \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \\ \vdots \\ \Sigma x_i^M y_i \end{bmatrix}$$

$$\boxed{(\phi^T\phi)^{-1}(\phi^T\phi)w = (\phi^T\phi)^{-1}\phi^T y \Rightarrow w \, (\phi^T\phi)^{-1}\phi^T y}$$

④ $(\phi^T\phi)w=\phi^T y$



$$(\phi^T\phi)w=\phi^T y \quad /\cdot(\phi^T\phi)^{-1}$$

$$\underbrace{(\phi^T\phi)^{-1}(\phi^T\phi)}_{I}w=(\phi^T\phi)^{-1}\phi^{\overset{I}{\cancel{T}}}y$$

$$\underline{Iw=(\phi^T\phi)^{-1}\phi^T y}$$

$$\Downarrow$$

$$\boxed{w=(\phi^T\phi)^{-1}\phi^T y}$$